GuitarTap Beta Release Notes

Version 1.0 · Build 244

What's New Since Build 180

New Features

Comparison Mode — Major Upgrade

- The Analysis Results panel now shows a **Comparison Results table** while comparing: Air, Top, and Back peak frequencies are displayed in a grid, one column per spectrum, with colour-coded dots matching the chart curves. Missing modes show "—".
- **Save a Comparison** — while a live comparison is active, the Save button stores the entire overlay as a single record in the Measurements list. Reloading it later restores all spectra, colours, and the Comparison table exactly as they were.
- **Export Comparison Spectrum** — Export Spectrum during a live comparison (or from a saved comparison record) produces an overlay PNG with all coloured curves and a legend.
- **Export Comparison PDF** — Export PDF Report during a live comparison produces a comparison PDF containing the spectrum overlay image and the Air / Top / Back frequency table. (Previously unavailable during comparison.)
- **Import Comparison** — saved comparison `.guitartap` files can be imported and reloaded.

Accept / Redo for Plate and Brace Measurements

- After each tap phase (Longitudinal, Cross-grain, FLC) the spectrum now **freezes automatically** and the detector pauses while you review the result.
- The **Pause button becomes Accept** (green) — tap it to lock in the result and advance to the next phase.
- The **Cancel button becomes Redo** (labelled "Redo L", "Redo C", or "Redo FLC") — tap it to discard the phase and re-tap without restarting the whole sequence.
- The live spectrum is suppressed during review so the captured result stays on screen.
- The phase progress indicator ("Phase 1/2", "Phase 1/3") is shown in the status bar; it is hidden when there is only one phase (Brace mode).

Pause / Resume Now Available in Guitar Mode

- Pause was previously only available during plate/brace sequences. It now works in guitar mode too, letting you freeze the live display, adjust Threshold and Peak Min, and re-arm without triggering an accidental measurement.

Edit Measurement Location and Notes

- Saved measurements can now have their location label and notes edited after the fact. Long-press (iOS) or right-click (macOS) a measurement row to access Edit Location & Notes.

Spectrum Chart Title

- The chart now displays the measurement's location label (or "New" if unsaved) as the chart title, making it easier to track which measurement is on screen.

Loaded-Measurement Warning

- When a saved measurement is loaded into the view, a warning is shown in the status bar if the measurement's recorded settings (Threshold, tap count) differ from the current live settings.

Improvements

Plate and Brace

- Adaptive noise floor for plate/brace live measurements — the detector now uses an EMA-based noise floor rather than a fixed threshold, reducing false triggers during the multi-phase sequence.
- FLC peak detection refined for better identification of the weaker FLC resonance.
- Peak selection in the Analysis Results panel for plate/brace is now cleaner — phase badges (L, C, FLC) are shown as informational labels, not interactive controls.
- Results ordering for plate and brace has been reorganised to match the natural measurement sequence.
- PDF output for plate/brace now matches what is shown in the live results panel.

Peak Frequencies

- All peak frequencies are now displayed to **0.1 Hz precision** (one decimal place) in the Results panel, Comparison table, and PDF reports.

Spectrum Image Export

- Exported spectrum images now match the live view: same chart title, same mode-line label positions, and correct coordinate handling for dragged annotations.

Measurements List

- Default documents folder changed to `Documents/GuitarTap` for cleaner file organisation.
- The last-used export/import directory is remembered between sessions.
- Duplicate measurement import is handled correctly.

macOS Menu Bar

- Menu bar items have been cleaned up and reorganised.
- The Edit menu no longer flickers on macOS.
- The Help menu item now works correctly again (was broken in an earlier build).

Show Unknown Peaks

- Fixed a bug where toggling "Show Unknown Modes" in Settings did not update the peak annotations already displayed on the chart.

Status Bar

- Status bar values (peak magnitude, input level) now update at the correct rate and show consistent numbers between the left and right sides.
- Status bar correctly shows different information for guitar vs. plate/brace measurements.

Quality Colour Indicators

- Fixed a duplication where the quality colour was being applied incorrectly in the PDF vs. the live results view.

Bug Fixes

- Fixed display oscillation (spectrum flickering between states) that could occur in certain conditions.
- Fixed import error handling — import failures now report correctly rather than silently failing.
- Fixed the Save button disappearing unexpectedly in some states.
- Fixed wrong coordinate space being used when exporting a spectrum with repositioned (dragged) annotations.

- Fixed scroll wheel on macOS causing too-rapid axis updates and visual clutter.
- Fixed cross-platform import: measurements saved on iOS can now be imported on macOS and vice versa, even if the microphone name doesn't match (a fallback is applied).

Build number is generated automatically from the git commit count.